ENGR 111
Introduction to Engineering
Class #5

Criteria, constraints and decision matrices

Choosing, and being able to say why

Where we are today

The design process has eight steps. Here we focus mostly on step 5.

Step 4. Development of alternative designs. Several different ways to solve the same problem.

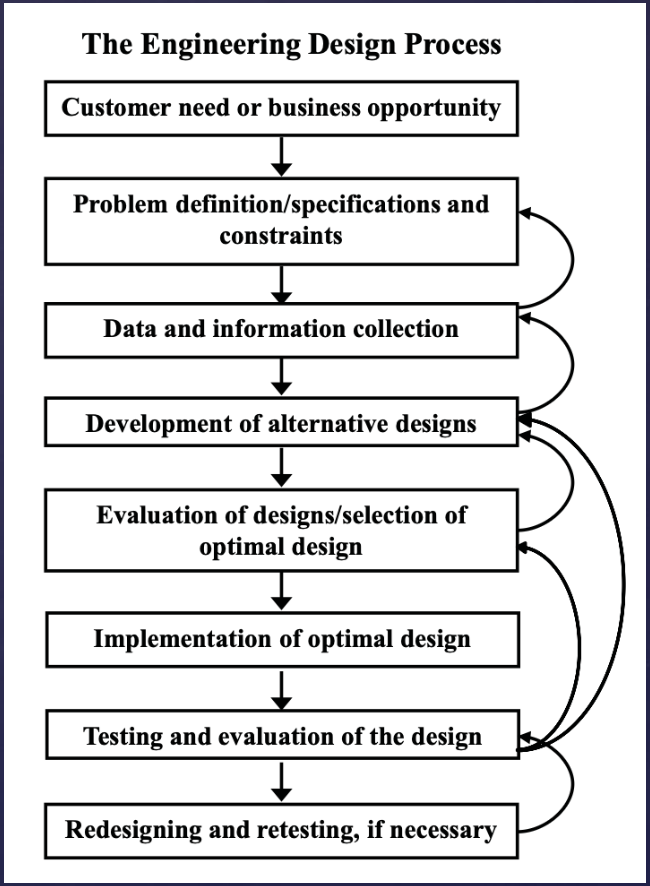
Step 5. Evaluation of designs, selection of the optimal design. Compare alternatives and pick one.

The textbook describes step 5 in one sentence.

"Based on a comparative evaluation, the optimal design will be selected."

Step 4

Step 5



Step 4 produces the candidates. Step 5 picks one of them.

Split the design into blocks

A robot that carries parcels across campus and into building lobbies.

Break the design into blocks. Each block is independent, or close to it, so you can work on one without holding the whole robot in your head. That is abstraction and is how teams can work independently on separate parts of a larger project.

Each block has its own set of options. Step 4 calls those the alternative designs.

Consider the drive system as a block. This describes the wheels or tracks, the motors that turn them, and the geometry that decides how the robot changes direction.

https://www.forbes.com/sites/christopherelliott/2019/02/03/what-are-the-rules-for-robots-delivering-food/

Chassis → **Drive system** → Power → Sensing and control → Payload box

Four drive configurations

There are more than four. These are four you decided are worth putting in the table.

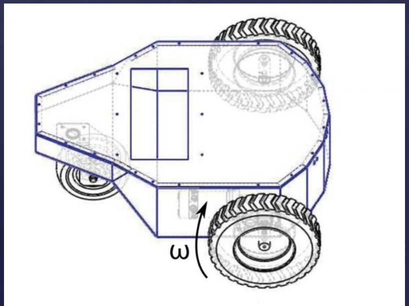
Selecting the drive configuration is a good early step since it impacts other blocks and the full shape of the robot.

1. Differential drive and a caster. Two driven wheels side by side and a caster holding up the rest. Drive them at different speeds to curve, or opposite ways to spin on the spot.

2. Four-wheel skid steer. Four driven wheels and no steering joint at all. It turns by driving one side faster than the other and letting the wheels slip.

3. Ackermann steering. Driven rear wheels, pivoting front wheels. What the car outside is doing.

4. Tracks. Two belts running over rollers. Spreads the load and grips loose or broken ground.



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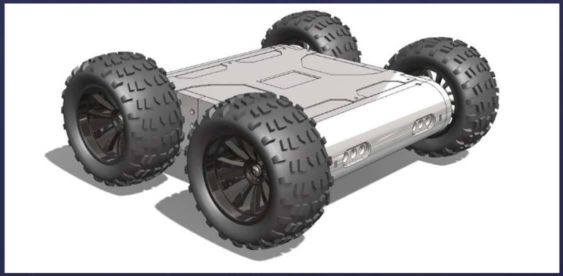

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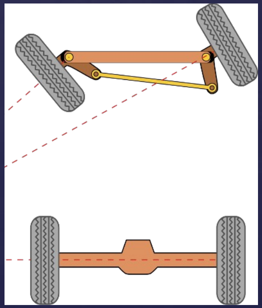

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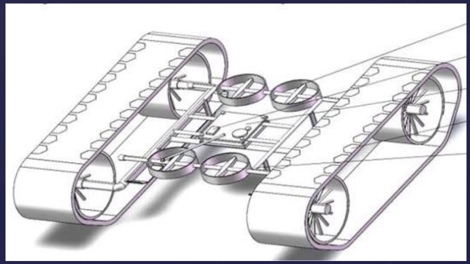

4.

The specifications, and who meets them

Suppose the team has decided to buy a drive base from a supplier and integrate it, not machine one themselves. That decision usually comes first, and it informs the columns below.

	Price ($)	**Turning circle (m)**	**Curb it climbs (cm)**	**Top speed (m/s)**	**Lead time (wk)**	**Meets spec**
must be	2500 or less	1.2 or less	3 or more	1.0 or more	8 or less	
because	the budget is $2,500	corridors are 1.2 m	the lobby lip is 3 cm	it keeps up on the sidewalk	the build slot opens in 8 weeks	
Differential drive + caster	900	0.9	3	1.8	2	**yes**
Four-wheel skid steer	1600	1.1	6	2.0	3	**yes**
Ackermann steering	1400	**2.4**	5	2.0	4	**NO**
Tracks	2200	1.0	12	1.2	8	**yes**

Ackermann needs 2.4 m to turn around and the corridor is 1.2 m, so it is out.

Price, curb and speed come back on the next slide as criteria. A number can screen and then rank.

Ranking what is left: Evaluating the tradeoffs

All three meet every specification. Criteria separate them on better or worse.

Design Weight	Price 25%	Lead time 15%	Curb it climbs 25%	Top speed 15%	Integration effort 20%	Total
Differential drive + caster	5	5	1	4	5	**385**
Four-wheel skid steer	3	4	3	5	2	**325**
Tracks	2	2	5	1	1	**240**

Differential drive wins on 385. Nothing wins every column (tradeoffs)

Note: integration is about the impact on other blocks. Skid steer and tracks slip in every turn, so the robot knows less about where it is, and you buy that back in sensors and compute.

Every number in that table is for one robot. Which of them move if you are building ten thousand?

Design your own base at that volume and the tooling gets paid off across the fleet, so price falls instead of rising. Lead time turns into a supply contract. Assembly time per unit starts to count, and somebody has to service robots nobody can drive out to.

Where did the scores come from?

Design Weight	Price 25%	Lead time 15%	Curb it climbs 25%	Top speed 15%	Integration effort 20%	Total
Differential drive + caster	5	5	1	4	5	**385**
Four-wheel skid steer	3	4	3	5	2	**325**
Tracks	2	2	5	1	1	**240**

A score is somebody's decision. A $900 base earned a 5 and a $2,200 base earned a 2. Another engineer could score those differently and defend it.

Keep the scale the same everywhere. 1 to 5, and 5 is always the good end, including for price, where low is better.

Say where each measurement came from. Measured on a prototype, read off a datasheet, estimated from something you know, or frankly guessed.

Mark the guesses. A table of guesses with a total along the bottom still looks authoritative. It is not.

Your textbook is honest about this. It says engineers bring common sense and experience to the design process, alongside the analysis and the modeling.

Where do the weights come from?

Design Weight	Price 25%	Lead time 15%	Curb it climbs 25%	Top speed 15%	Integration effort 20%	Total
Differential drive + caster	5	5	1	4	5	**385**
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Where weights come from. Experience, the customer, the market, the regulator, and whatever went wrong on the last project. They add to 100, so you cannot call everything important.

Everyone weights on their own, then average. It keeps the loudest person in the room from setting the priorities alone. Common practice on design teams.

Revising a weight is normal. Somebody walks the route and finds curbs on it. Curb goes from 25 to 45, price drops to 5, and skid steer wins on 325 instead. The process is iterative, and your textbook says so outright.

Rigging it is a different thing. Don't decide the answer first and then tune the weights until the table agrees.

One weight can flip the answer. So that weight is the real decision which you should carefully look at.

What the table leaves out

The matrix compared one block. The robot is not one block.

Blocks are coupled. A heavier drive needs a bigger battery and a stronger chassis, and the payload box shrinks to make room for both. The drive system table never sees any of that.

Only what you wrote down exists. A capability nobody entered as a criterion counts for nothing, because as far as the table is concerned it is not there.

The winner can still be wrong. If the answer feels wrong, either a weight is wrong or your instinct is. Find out which before you overrule it.

The table does not decide. You decide. What it does is make your reasoning visible enough that somebody else can argue with it.

Next steps (two possibilities):

1. Run the same procedure on the same block again

Choose between different differential drive/caster systems on the market (more tradeoffs).

2. Now run the same procedure on a different block.

Order matters. Now that the base has been chosen this drives requirements for how it talks to the controller and sensors, etc.